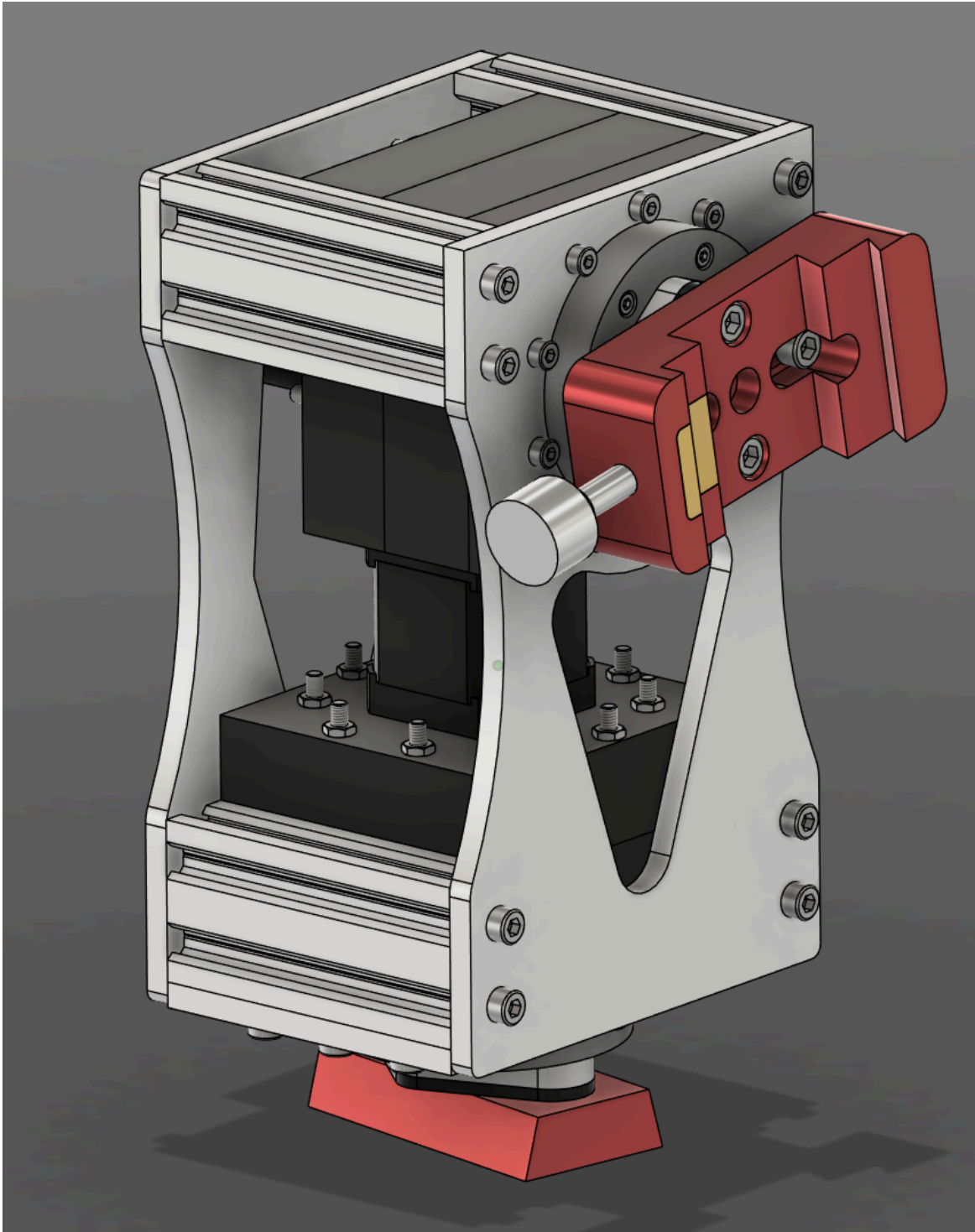


Assembly Instructions for the SpaceMount Equatorial Mount



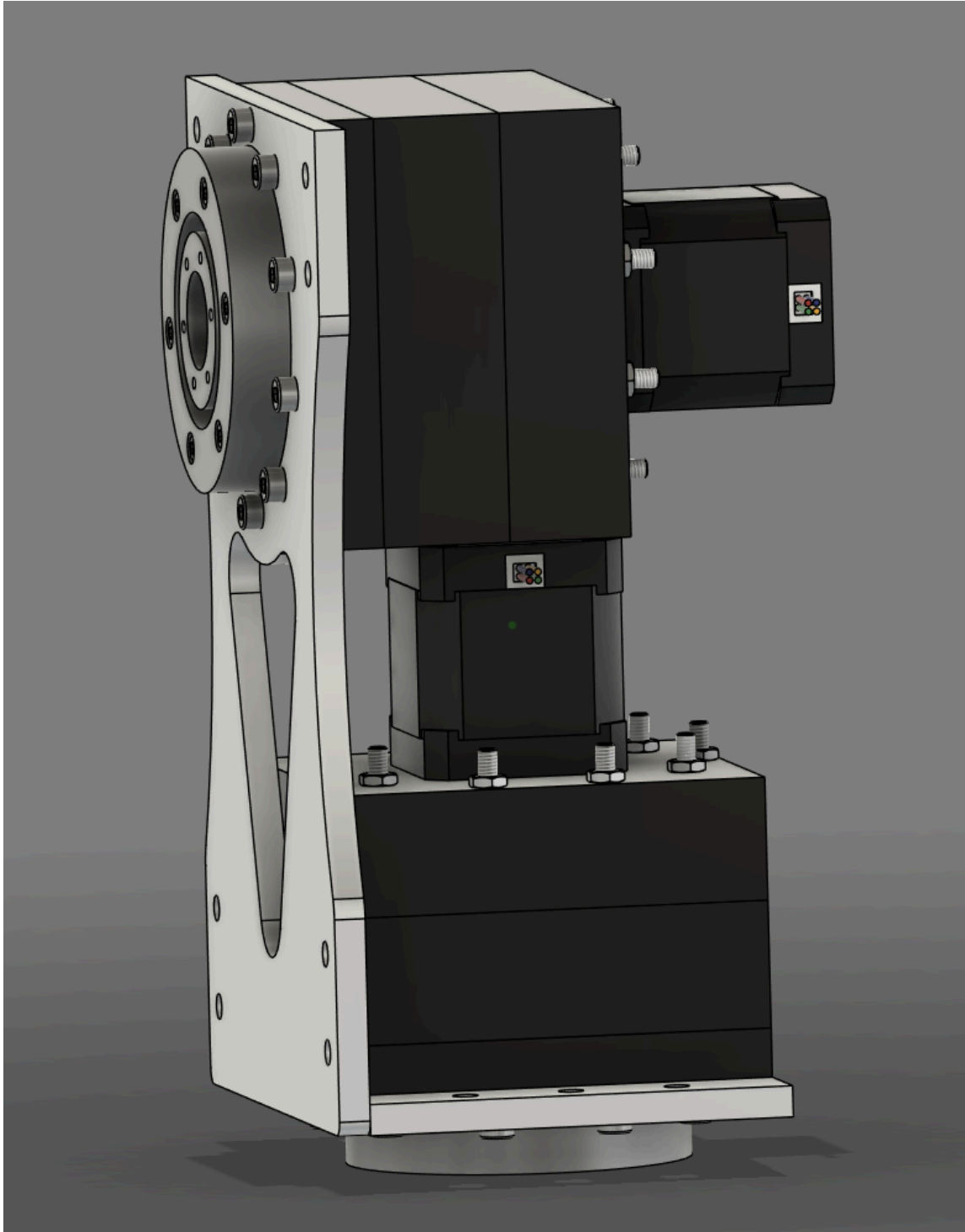
Preface

This is the assembly guide for my motorized equatorial mount that is part of my [SpaceMount](#) project. It is certainly not error free, but it is written to the best of my abilities. Always refer to the master CAD assembly provided in the Github Repository for confirmation of assembly. Best of Luck!!

Note: Unless otherwise stated, assume screws are Socket Head Cap Screws (SHCS).

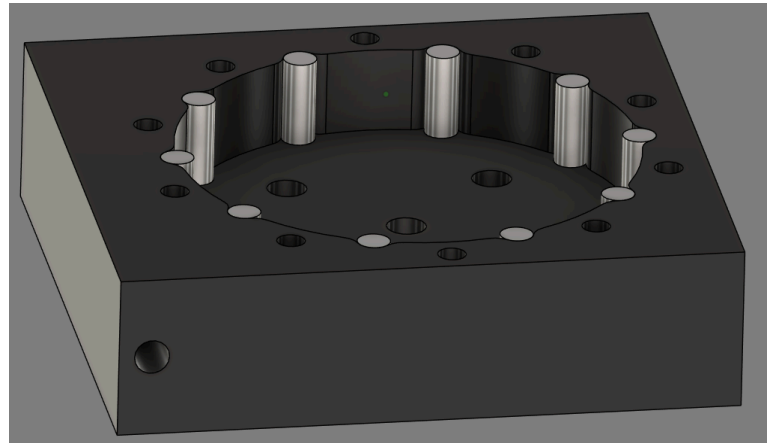
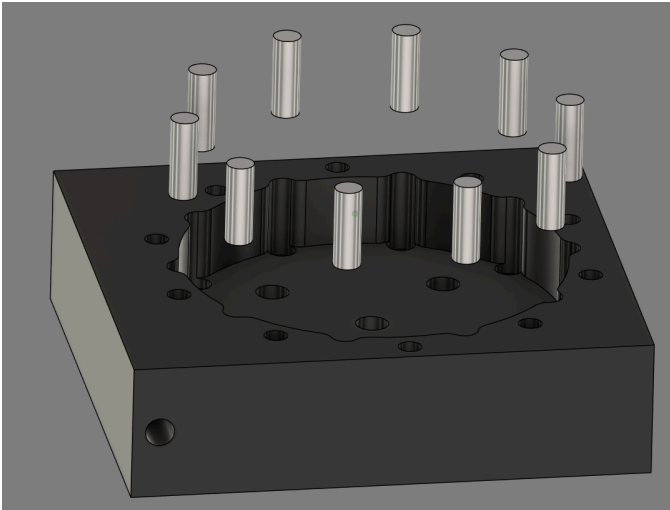
Part 1: The R.A. and Dec Cycloidal Gearboxes

Note: You may use the M4x75 bolts inserted from the rear to hold the gearbox together during assembly, but keep in mind they must ultimately be inserted from the front to be assembleable.

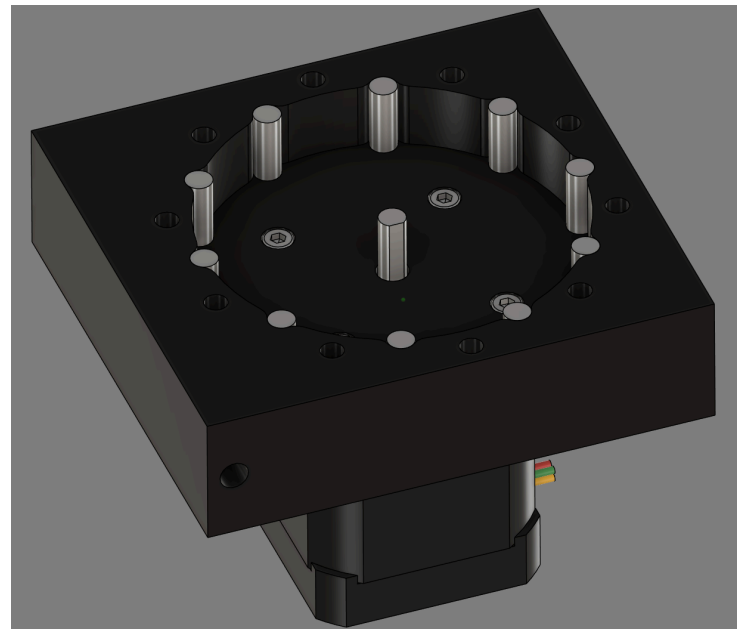
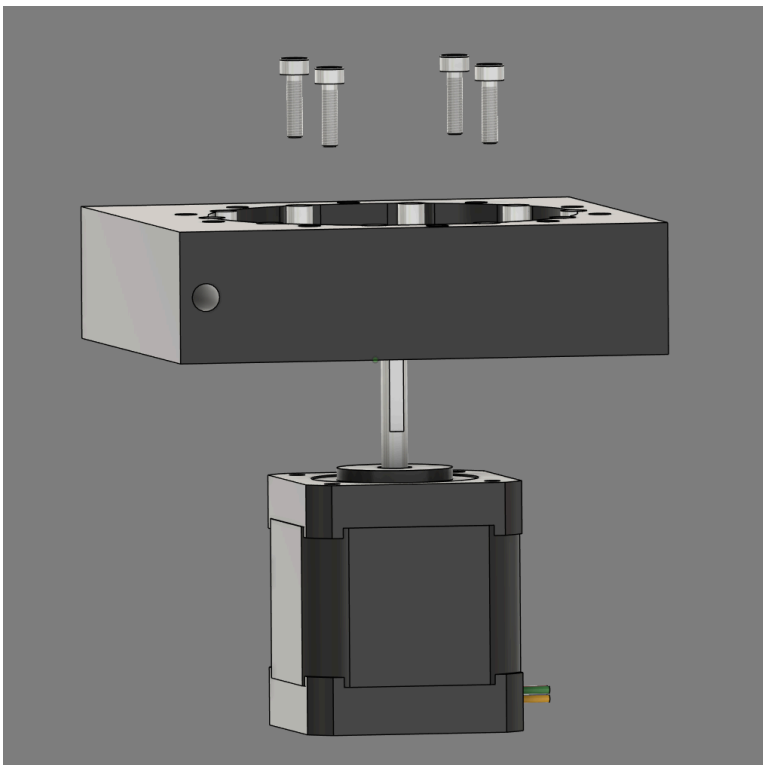


Step 1: Install Pins and Motor into Lower Housing

- Press 10x 5mmx15mm pins into the Lower Housing. May use a vice, arbor press, gentle hammering, etc.

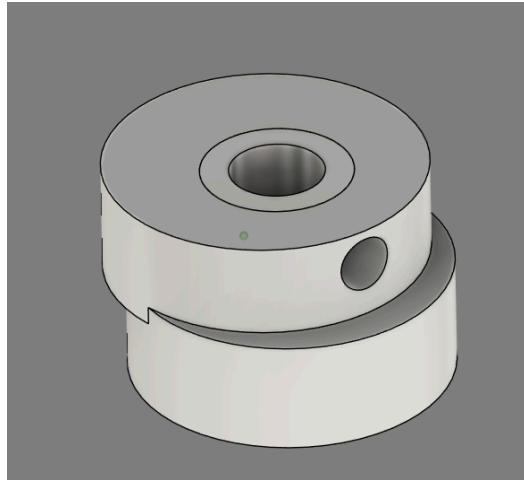
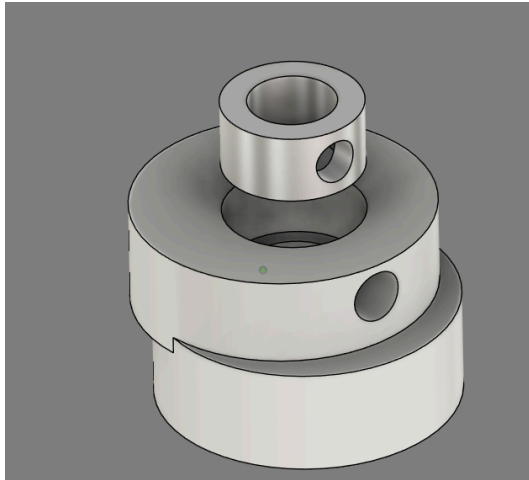


- Bolt 1x Nema 17 Stepper Motor into the bottom of the Lower Housing with 4x M3x12 Bolts.

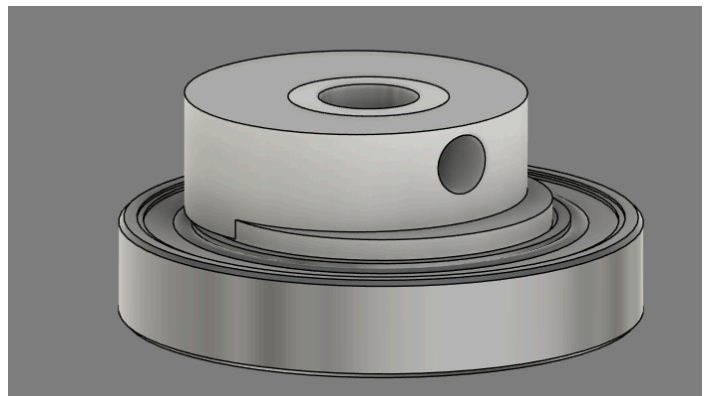
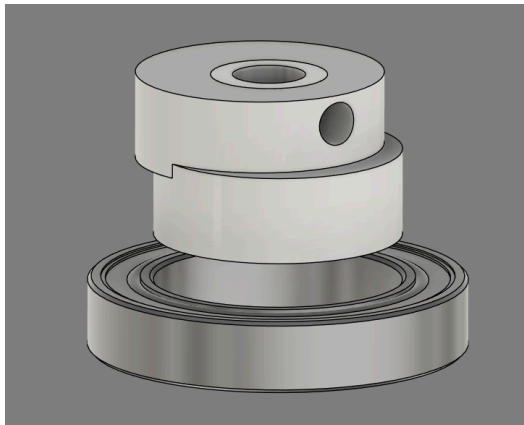


Step 2: Cam Assembly

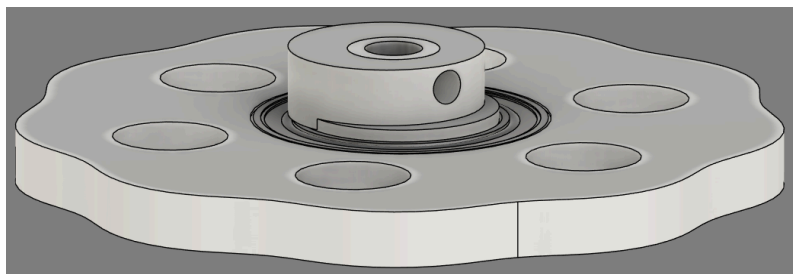
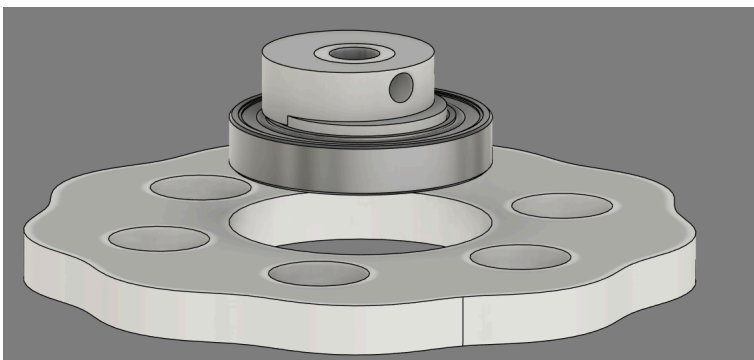
- Press the Shaft Collar into the Cam, taking care to align the setscrew hole with the hole in the Cam.



- Install 1x 6803 Bearing onto the Cam on the side without the set screw hole.

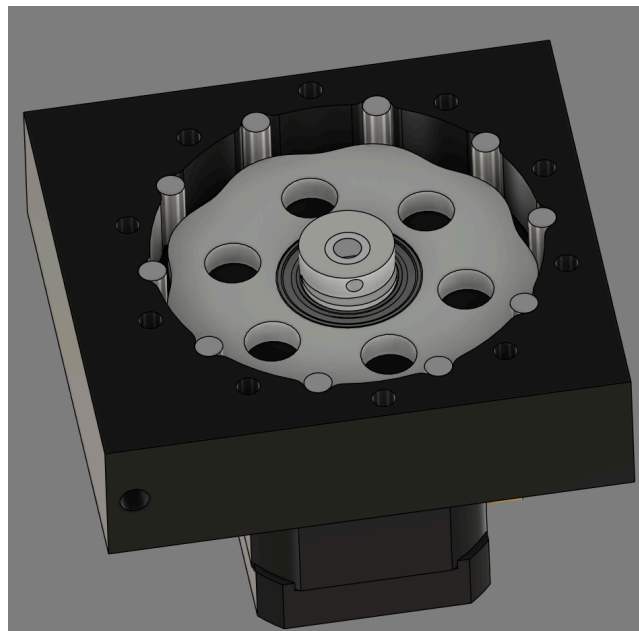
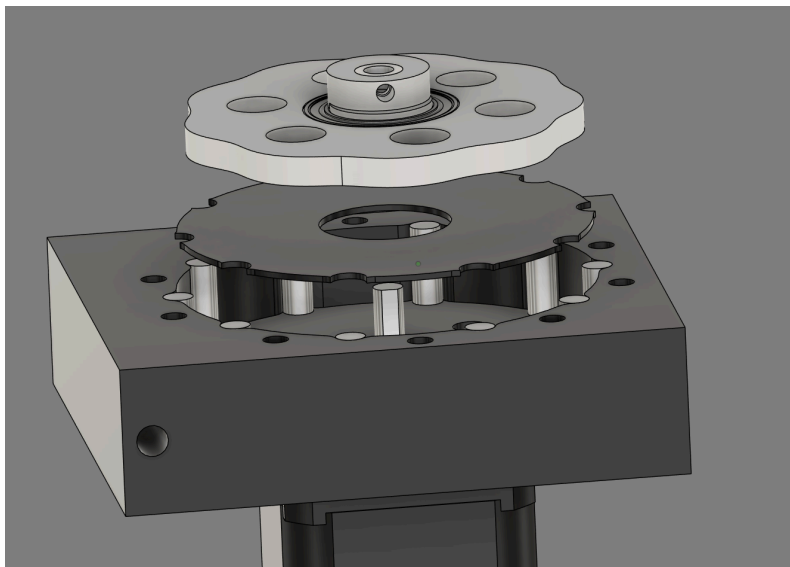


- Install 1x Cycloidal Gear onto the installed 6803 Bearing

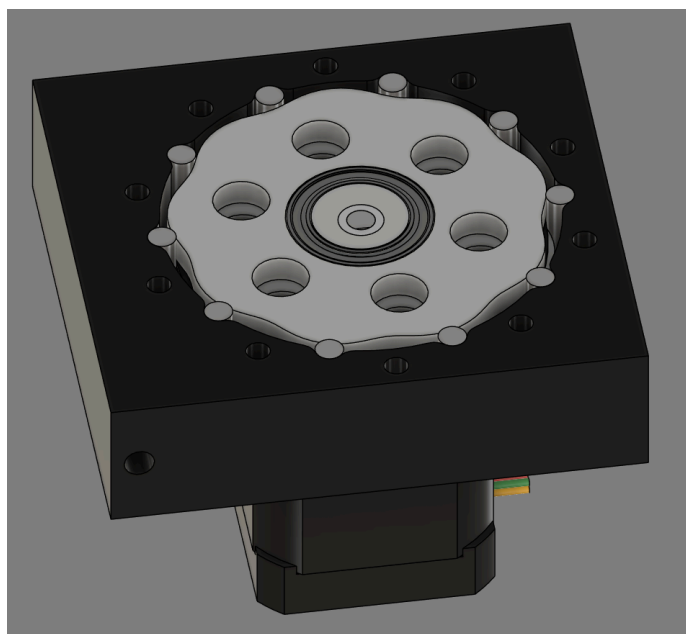
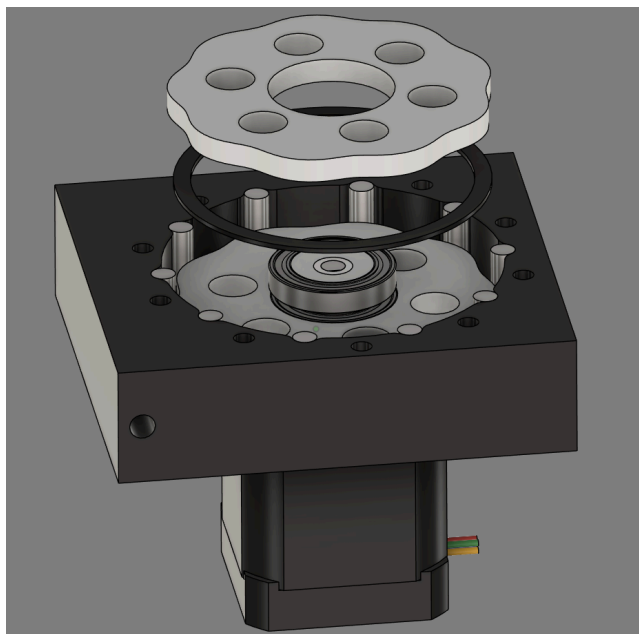


Step 2 Continued:

- Install the Cam Assembly onto the Nema 17 Stepper Motor Shaft and tighten the set screw. Make sure to insert the Bottom 1mm Spacer inbetween the Lower Housing and Cam Assembly.

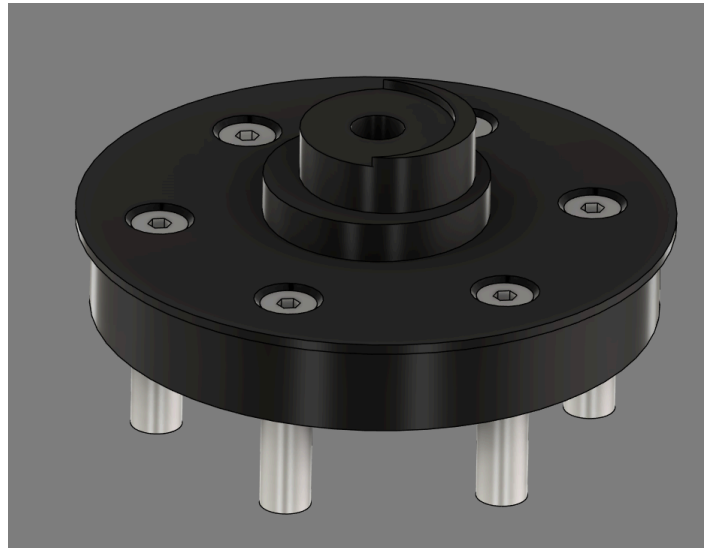


- Install the second 6803 Bearing and Cycloidal Gear onto the upper 6803 Bearing with a Mid 1mm Spacer between the two Cycloidal Gears. **Make Sure The Set Screw Is Tightened.**



Step 3: Internal Output Assembly

- Install 6x M3 15mm Standoffs and 6x M3x16mm Countersunk Bolts into the Lower Internal Output Shaft. The standoffs will likely need pressed into the holes on the underside.

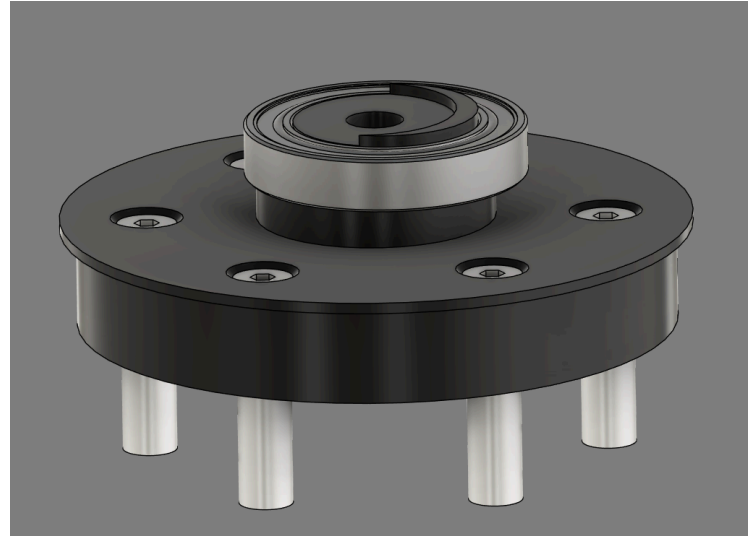
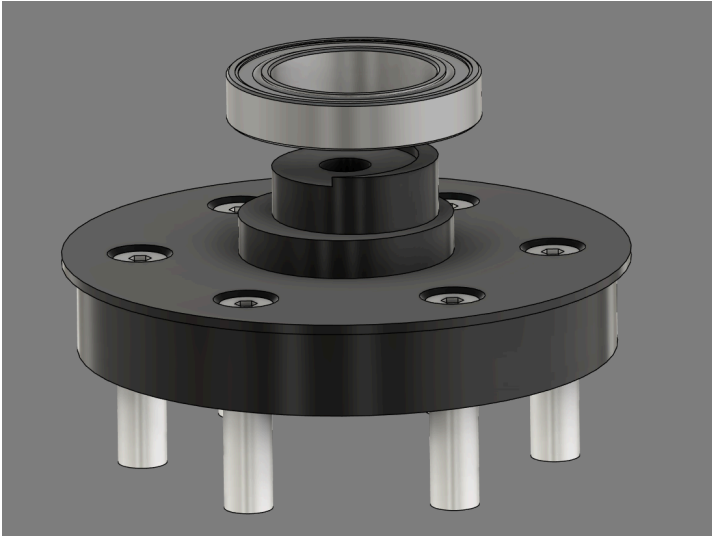


- At this time also install the M5 nut into the underside of the Lower Output Shaft. It should press into place, but feel free to add superglue if the fit is loose. This must be secured in place.

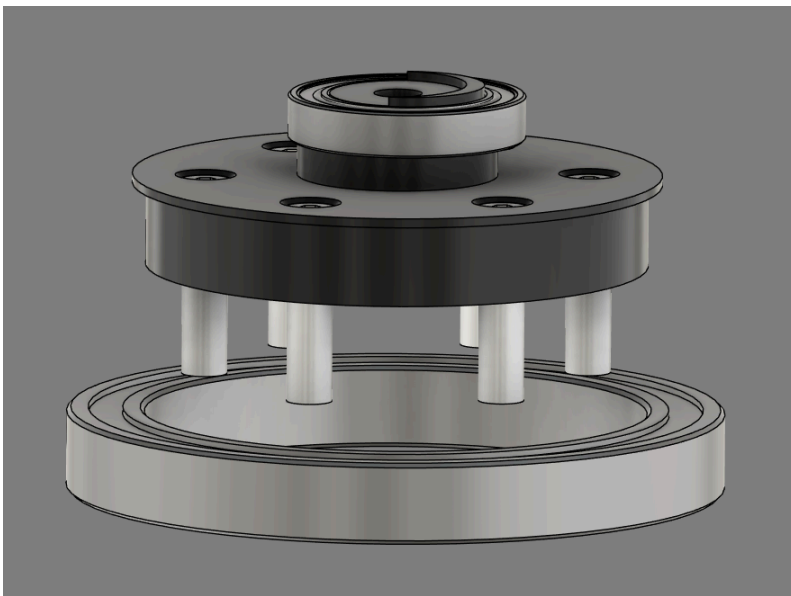


Step 3 Continued

- Install 1x 6803 Bearing onto the Lower Internal Output Shaft.

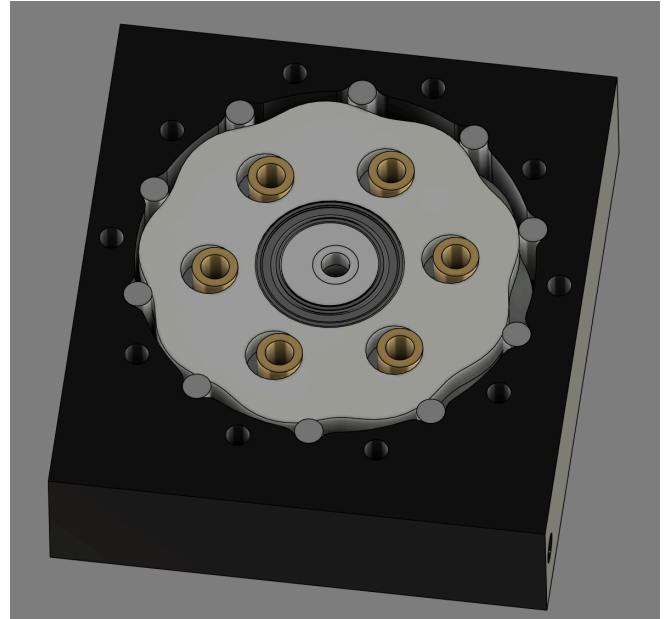
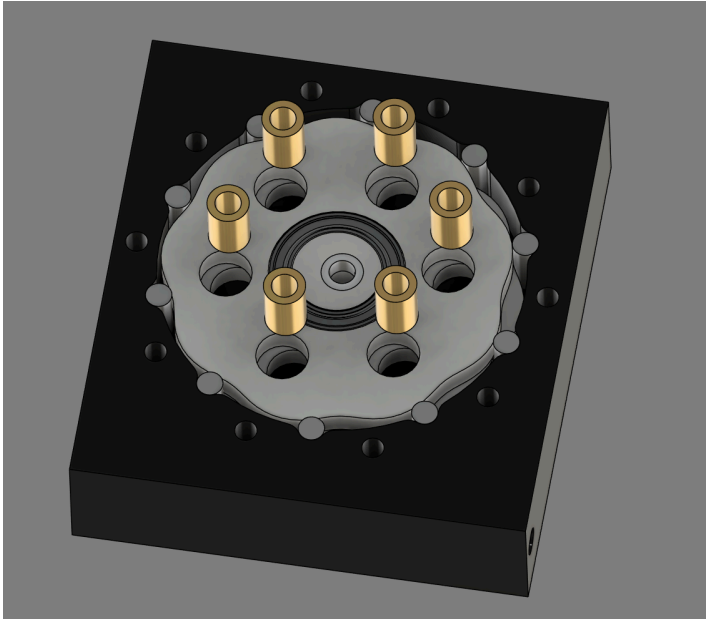


- Install the Lower Internal Output Shaft into the 6811 Bearing.

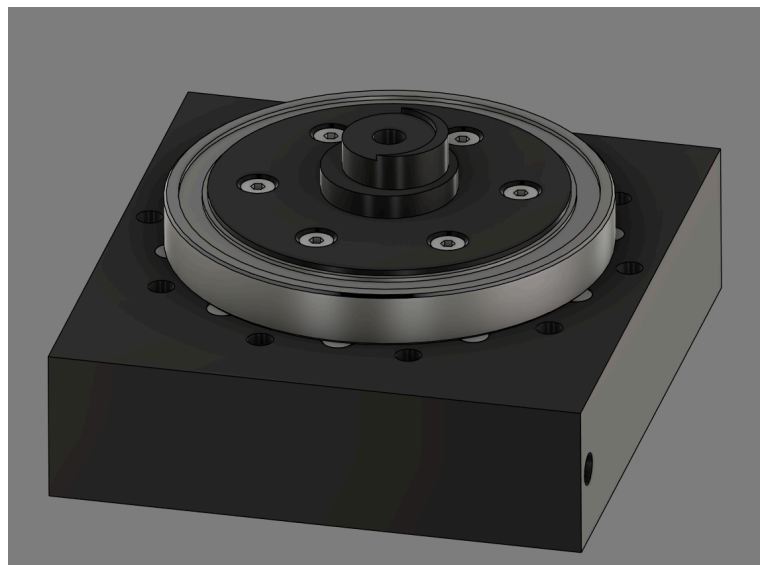
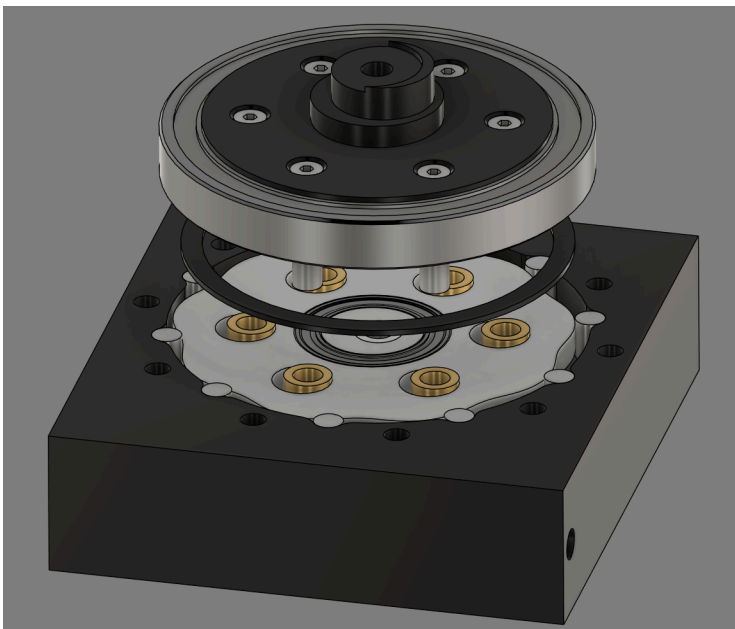


Step 4: Finishing off the First Gearbox Stage

- Place 1x Brass Bushing into each of the Cycloidal Gear Holes. These should pass through both gears and total 6x bushings.

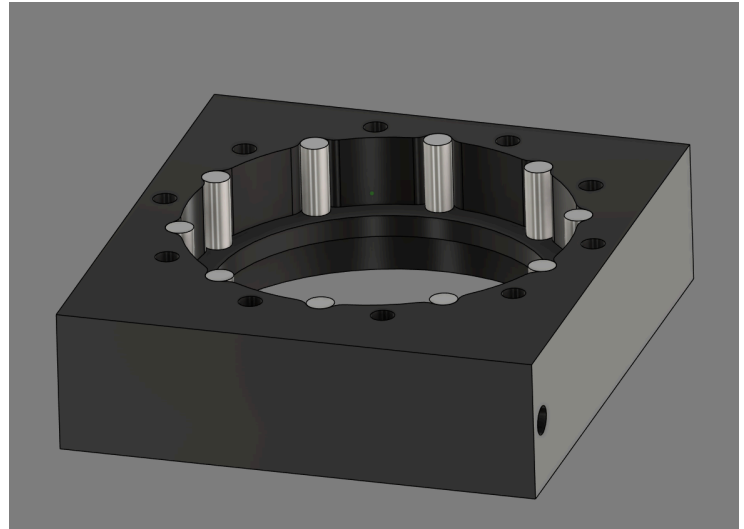
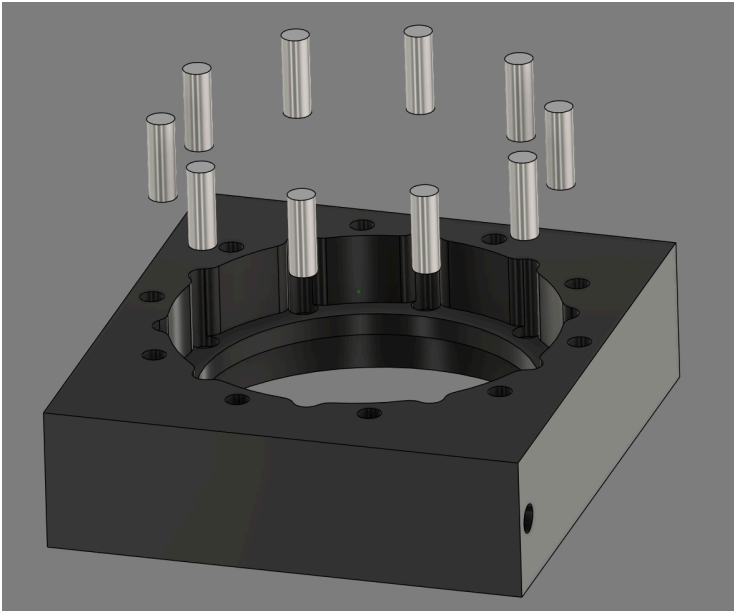


- Insert the aluminum standoffs on the Internal Output Assembly into the Brass Bushings. This will be a bit tricky to align. Place a 1mm Mid Spacer on top of the Upper Cycloidal Gear. This finishes the First Stage.

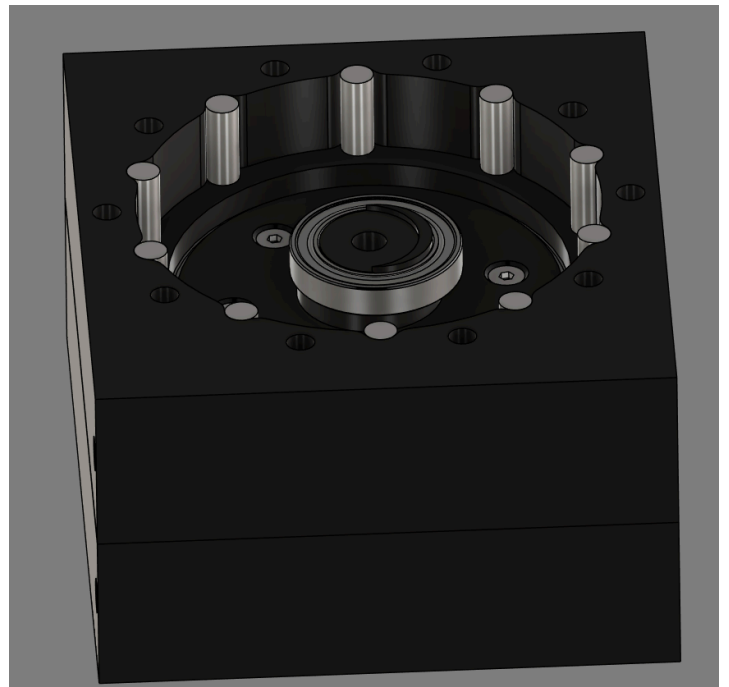
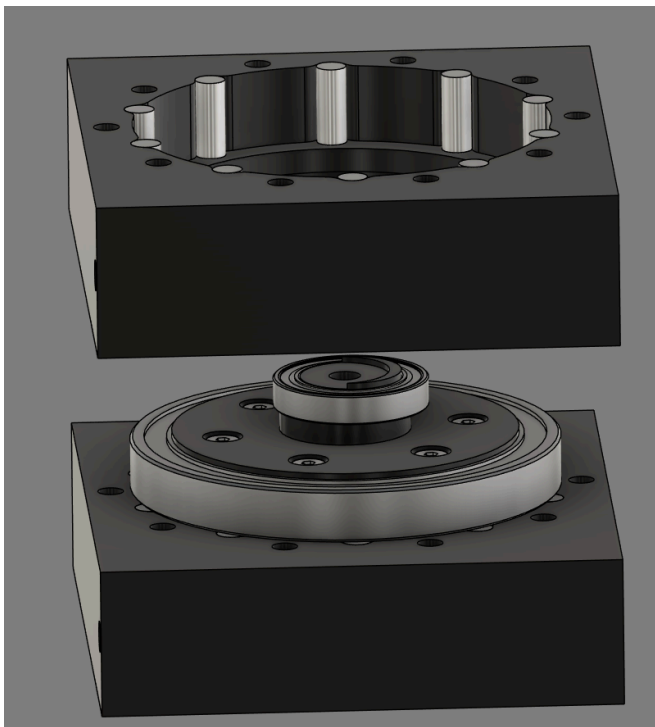


Step 5: Mid Housing

- Press 10x 5mmx15mm pins into the Mid Housing . May use a vice, arbor press, gentle hammering, etc.

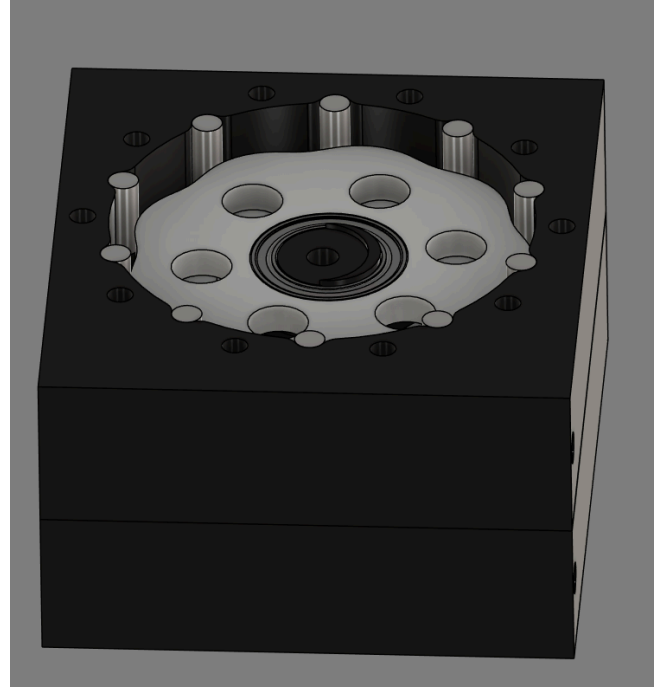
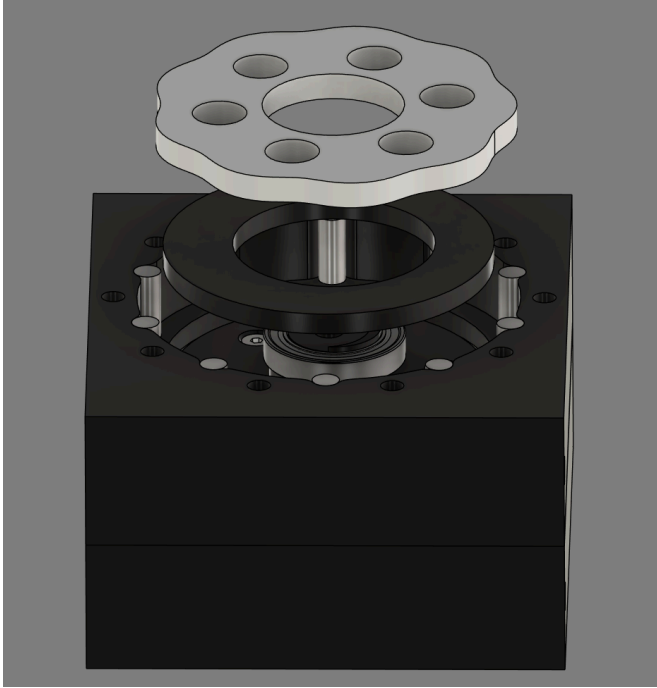


- Place the Mid Housing, now with pins inserted, onto the 6811 bearing. Keep the horizontal through holes aligned to the same faces.

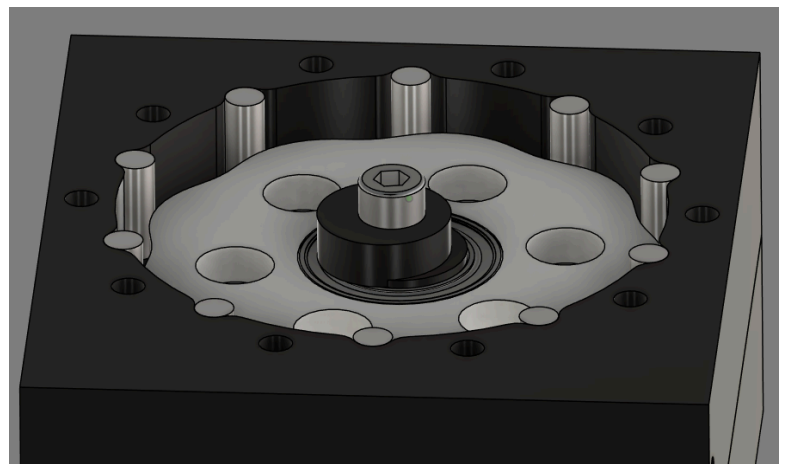
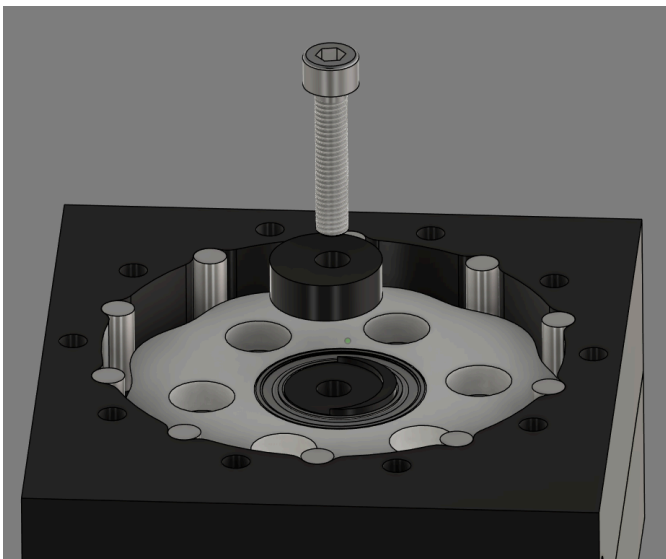


Step 6: Upper Stage Gearing

- Install the 4mm Mid Spacer on top of the Internal Output Assembly and then a Cycloidal Gear on top of that, onto the 6803 bearing.

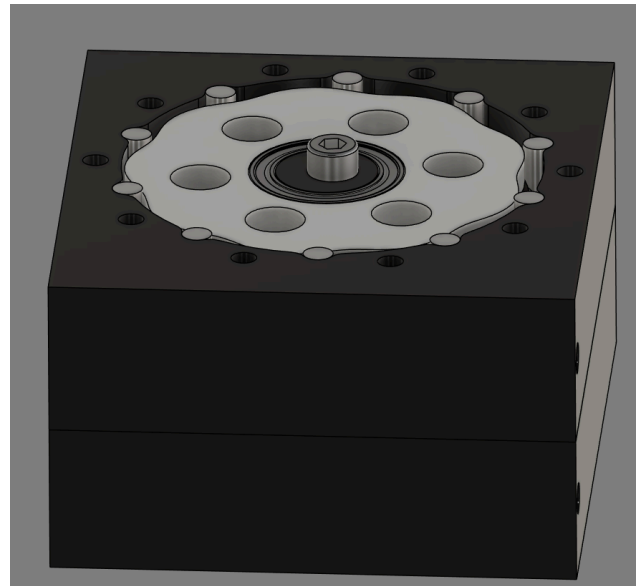
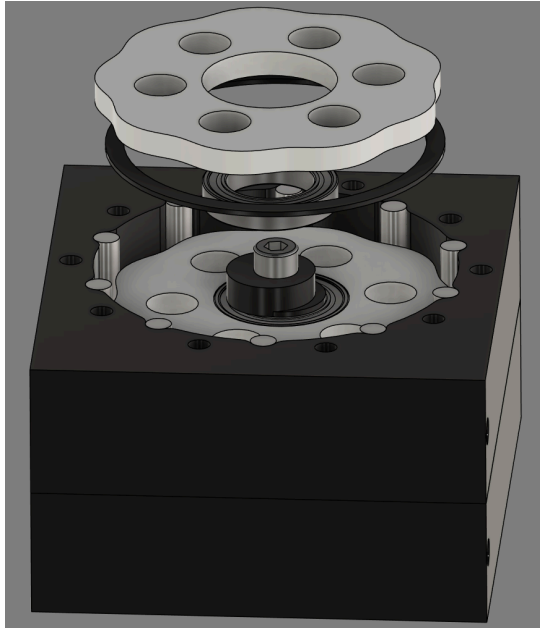


- Install the Upper Internal Output Shaft using 1x M5x25mm bolt. **Tighten this hard.** This compression provides structure to the internal output shaft and is highly important.

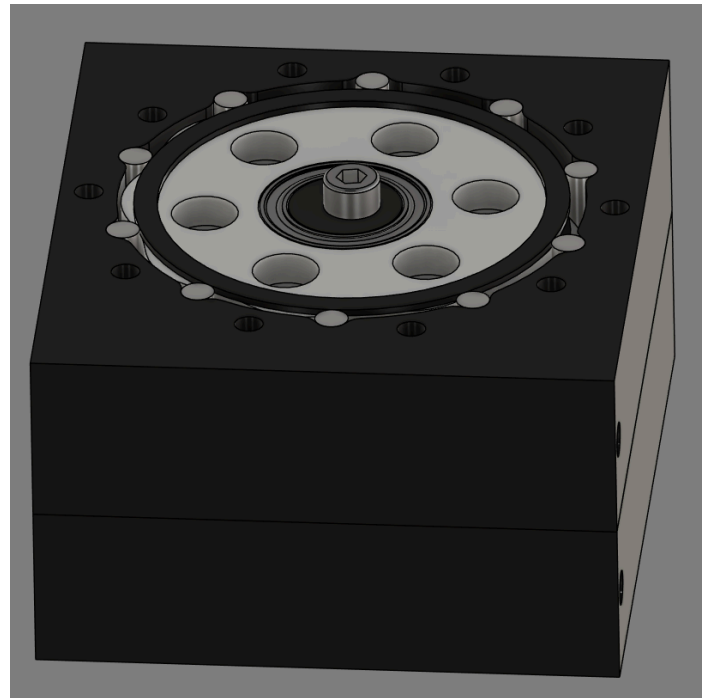
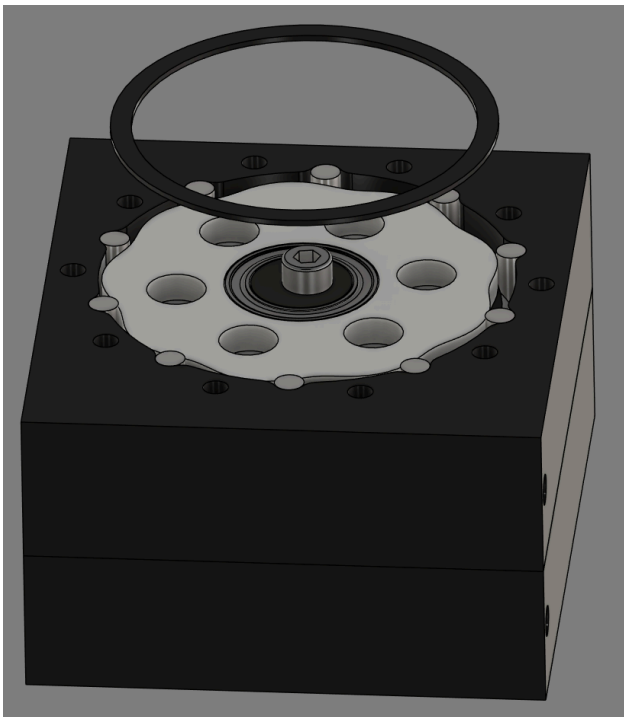


Step 6 Continued

- Install the second Cycloidal Gear, 6803 Bearing, and 1mm Mid Spacer. The spacer goes between the two Cycloidal Gears.

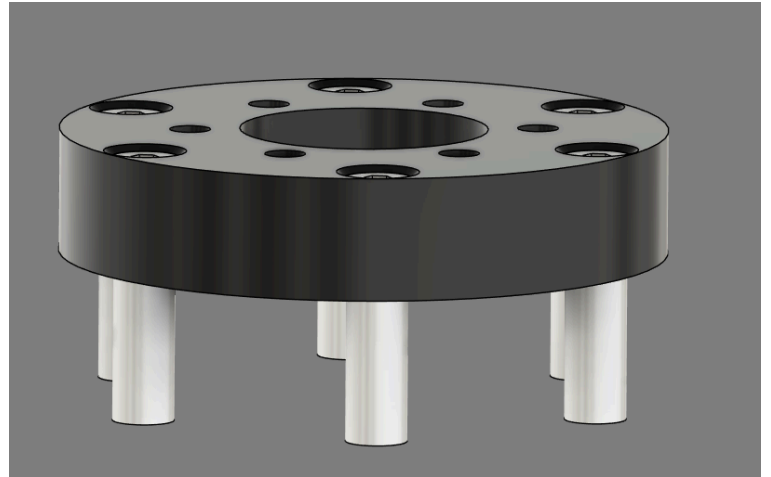


- Install one more 1mm Mid Spacer on top of the upper Cycloidal Gear.

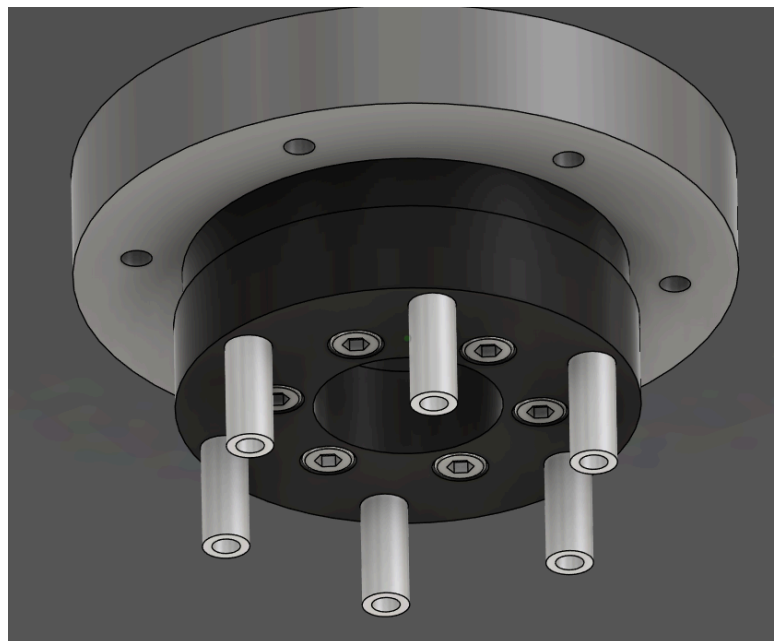


Step 7: External Output Assembly.

- Install 6x M3 15mm Standoffs and 6x M3x16mm Countersunk Bolts into the External Output Shaft. The standoffs will likely need pressed into the holes on the underside.

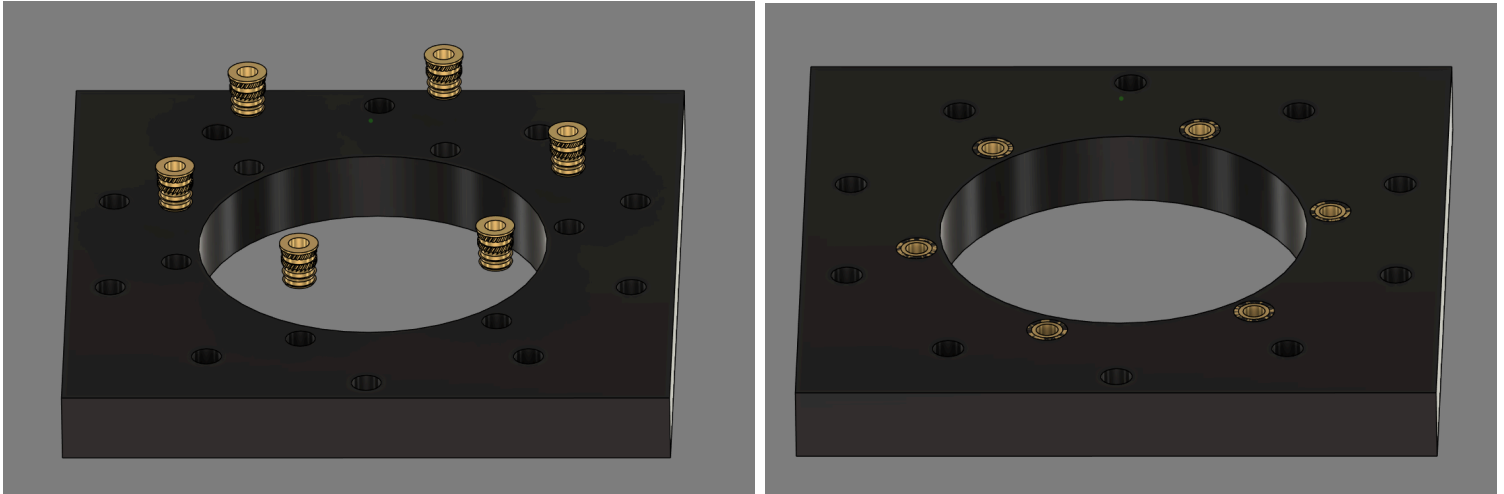


- Bolt the External Output Shaft to the RU42UCC0 Bearing using 6x M3x20mm Bolts. Place the External Output Plate between the External Output Shaft and the RU42UCC0. It will be sandwiched by the two parts.

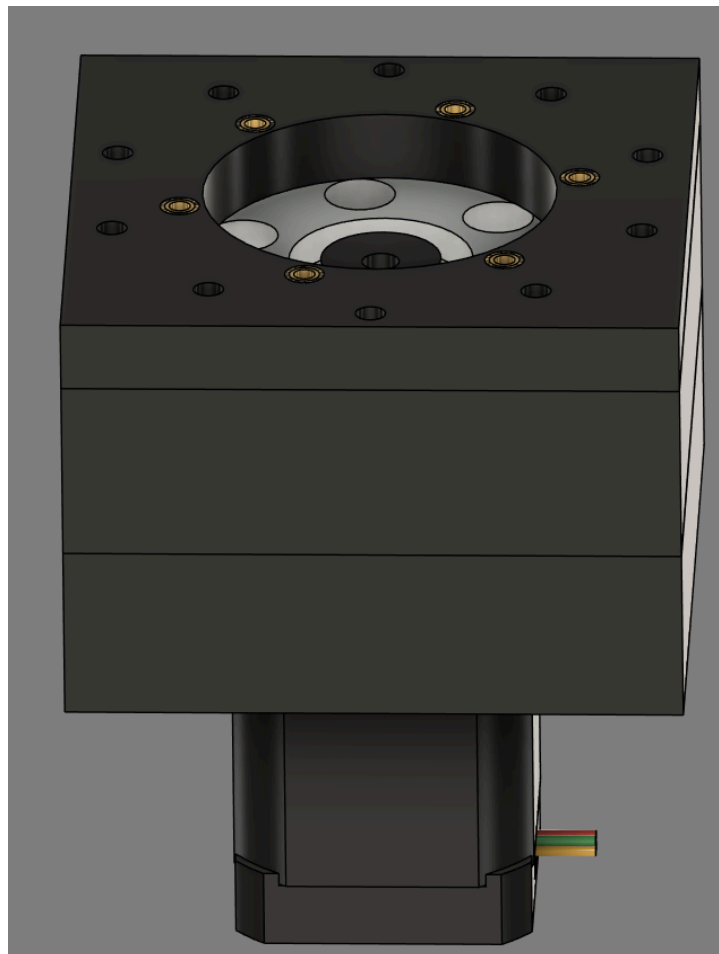


Step 8: Upper Housing

- Insert 6x Long M3 Heat Inserts into the Upper Housing.

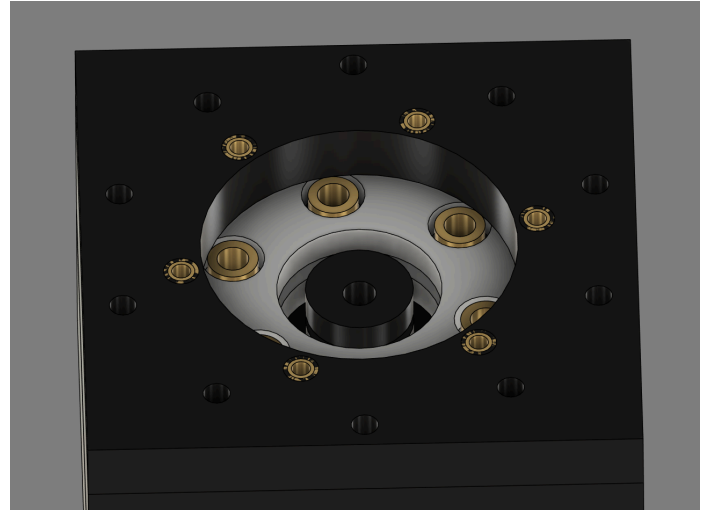
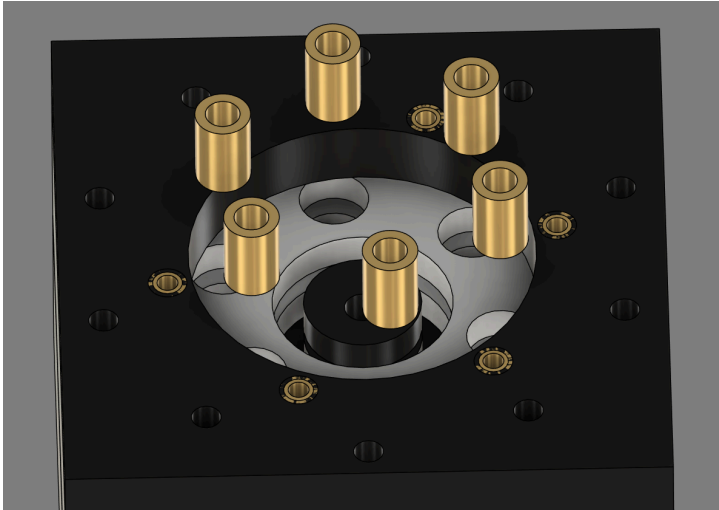


- Place the upper housing onto the gearbox stack.

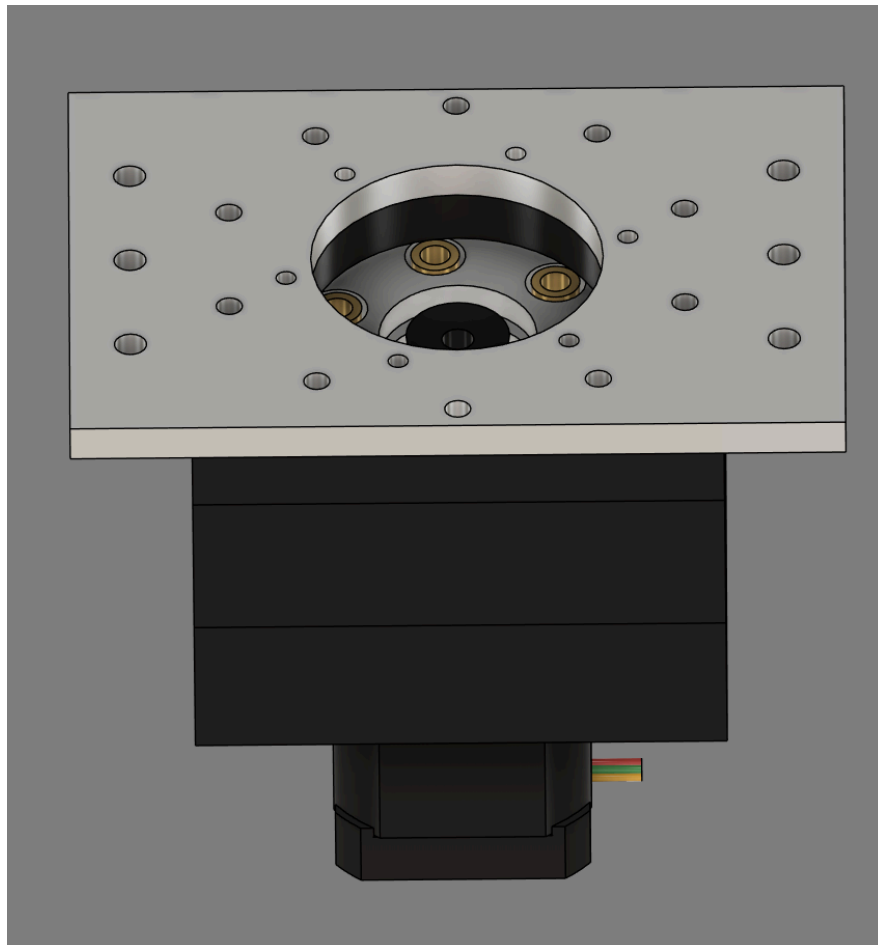


Step 9: Final Assembly

- Place 1x Brass Bushing into each of the Cycloidal Gear Holes. These should pass through both gears and total 6x bushings.

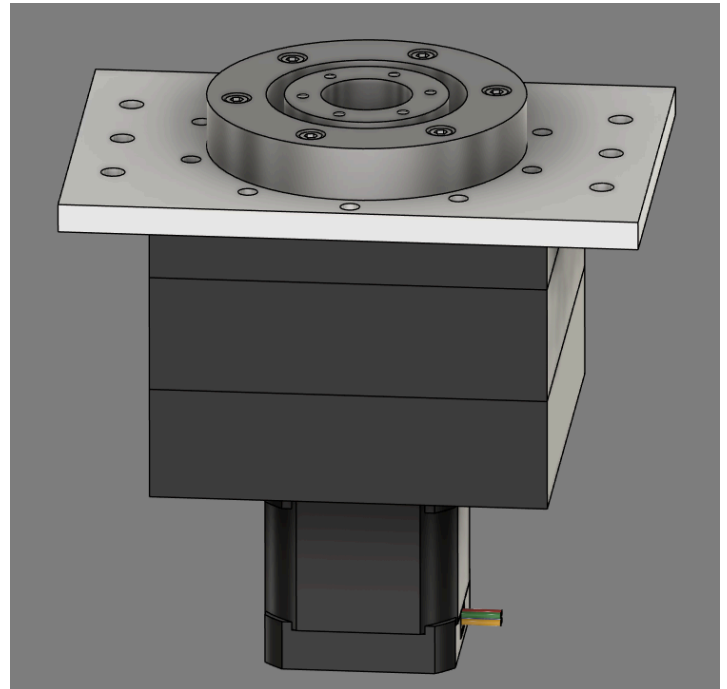
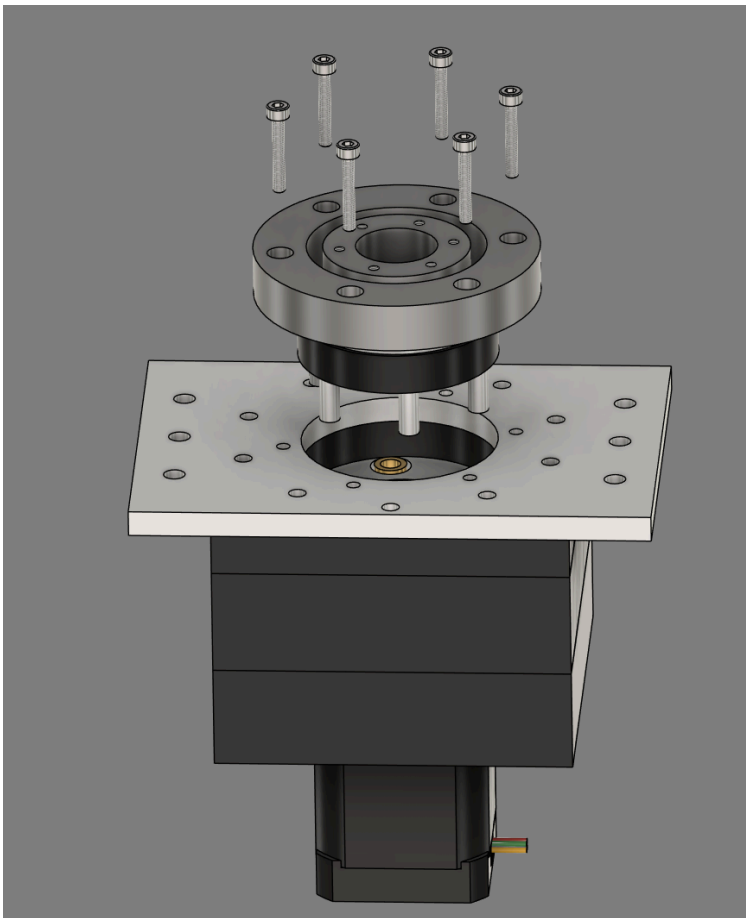


- Place the Bottom Plate onto the Gearbox Stack



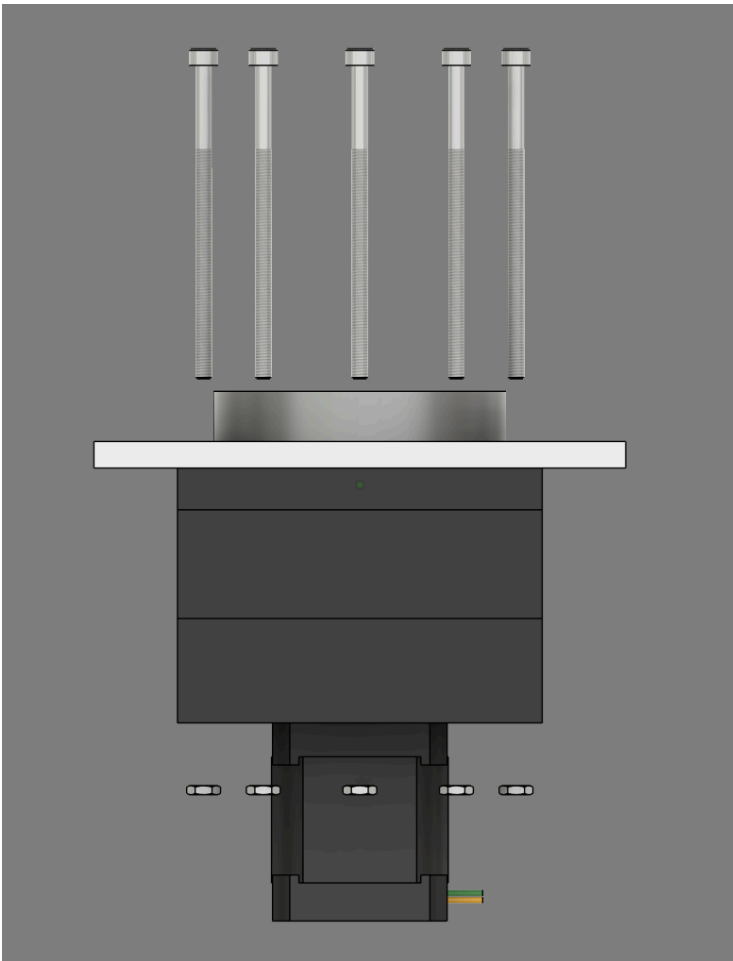
Step 9 Continued

- Insert the aluminum standoffs on the External Output Assembly into the Brass Bushings. This will be a bit tricky to align. Using 6x M3x25mm Bolts, attach the outer ring of the RU42UCC0 to the Gearbox Stack. These bolts go into the M3 Heat Inserts that are on the Upper Housing, bolting through the Bottom Main Plate.



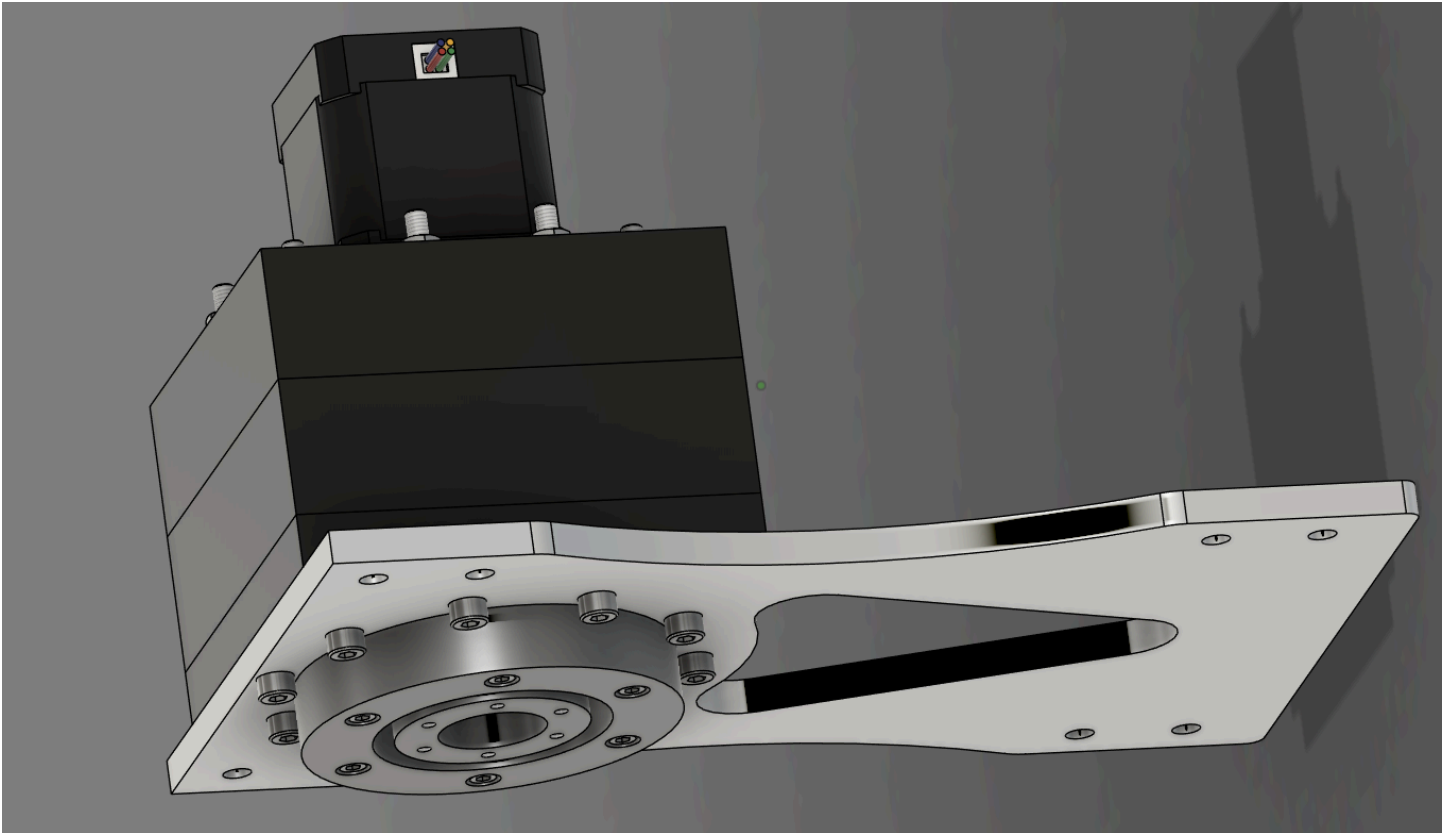
Step 10: Full Throughbolts and Finished

- Utilizing 10x M4x75 Bolts and 10x M4 Nuts, bolt through the entire gearbox assembly, from the Bottom Main Plate to the Lower Housing. With that, the R.A. Cycloidal Gearbox is finished.

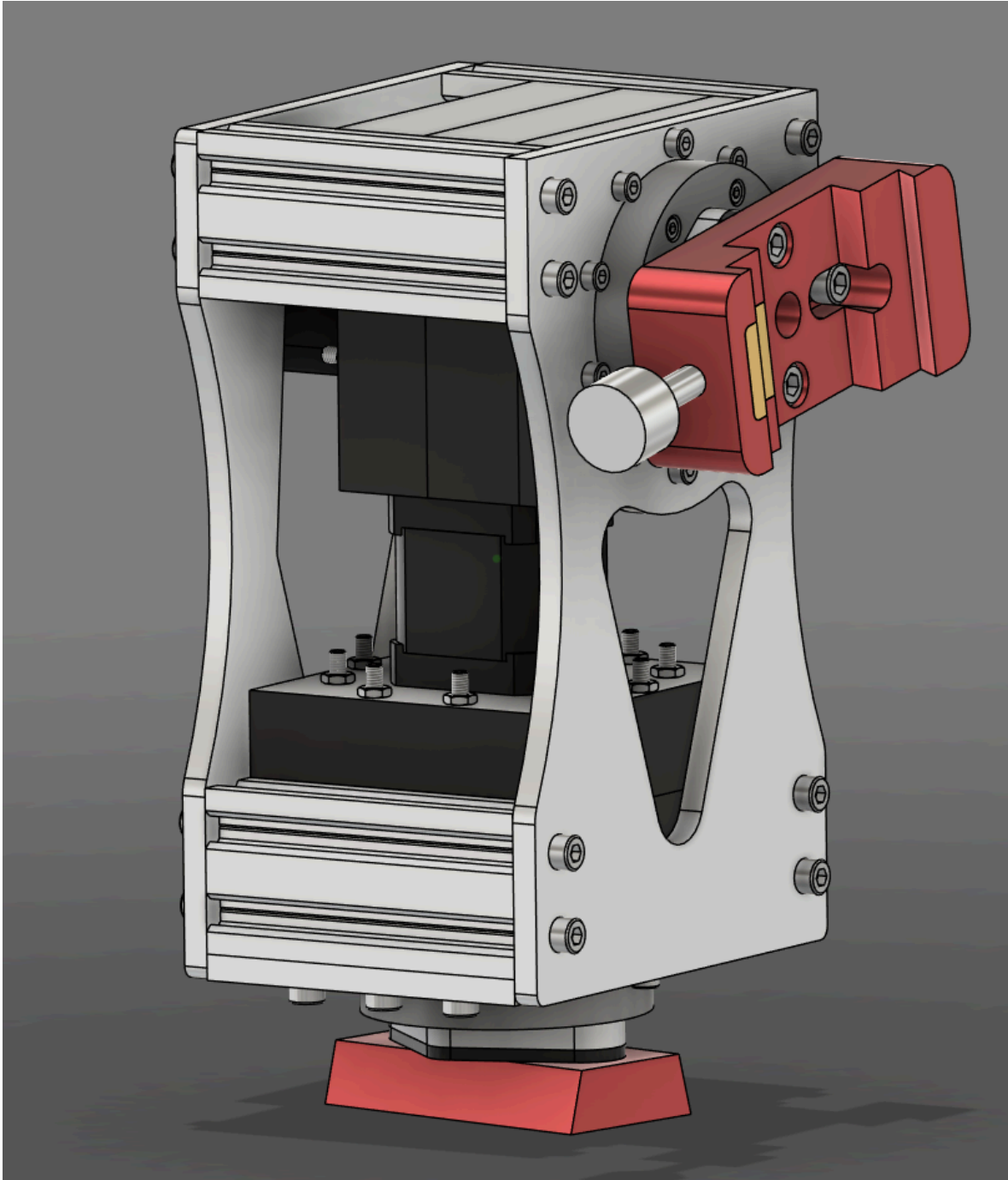


Step 11: Dec Gearbox

- Repeat steps 1-10 for the Dec gearbox, but substitute the Bottom Plate for the Front Plate. It should look like the following when complete:

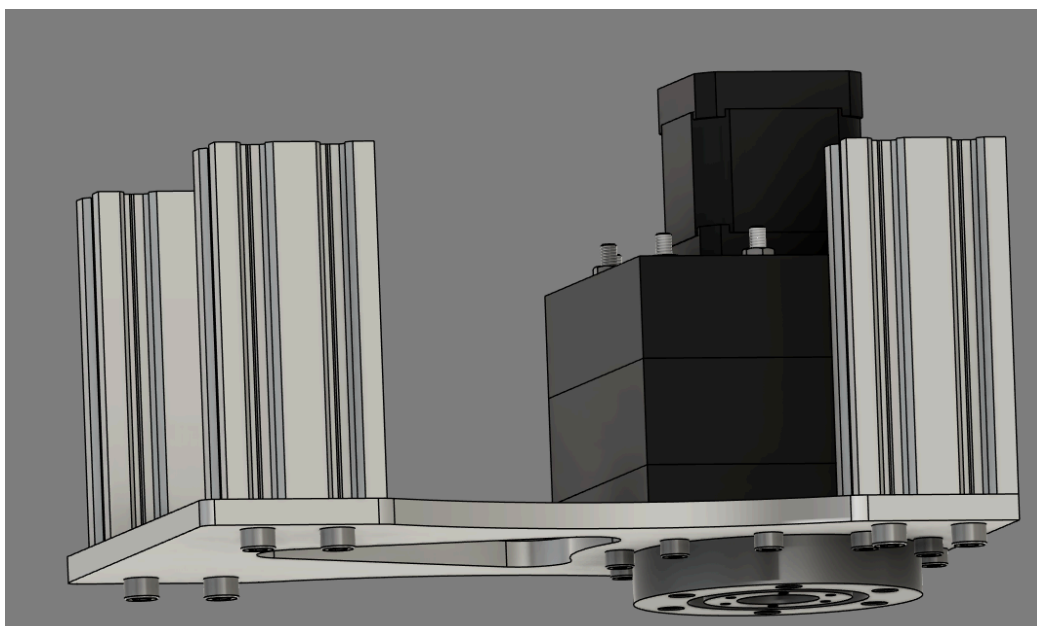
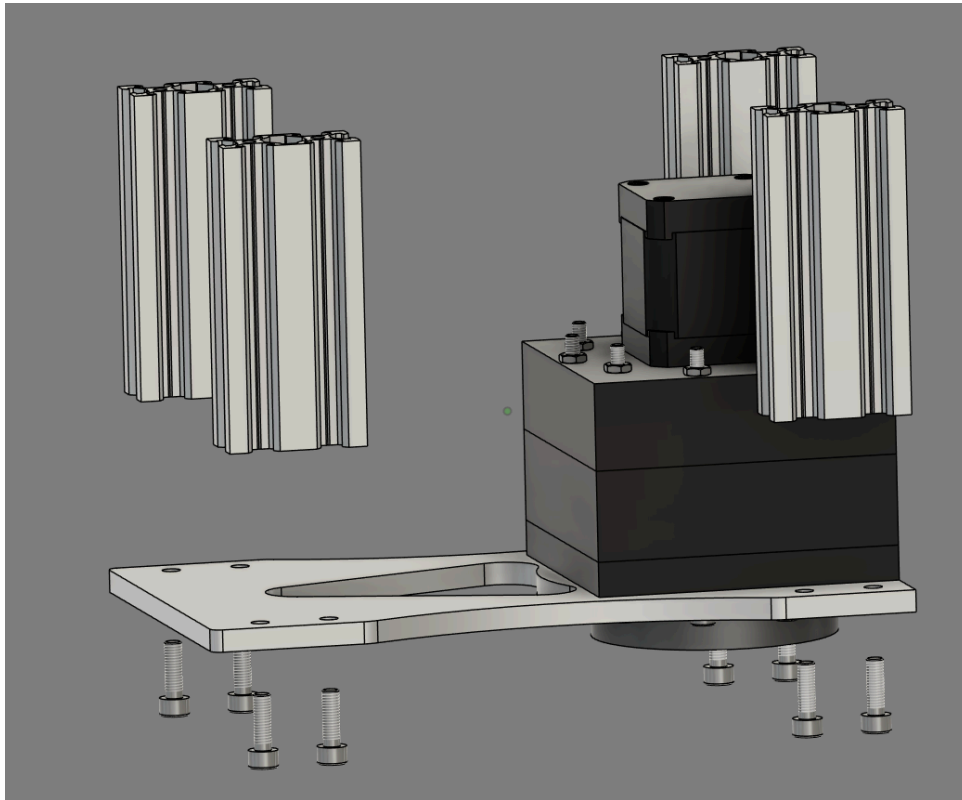


Part 2: Frame and Final Assembly



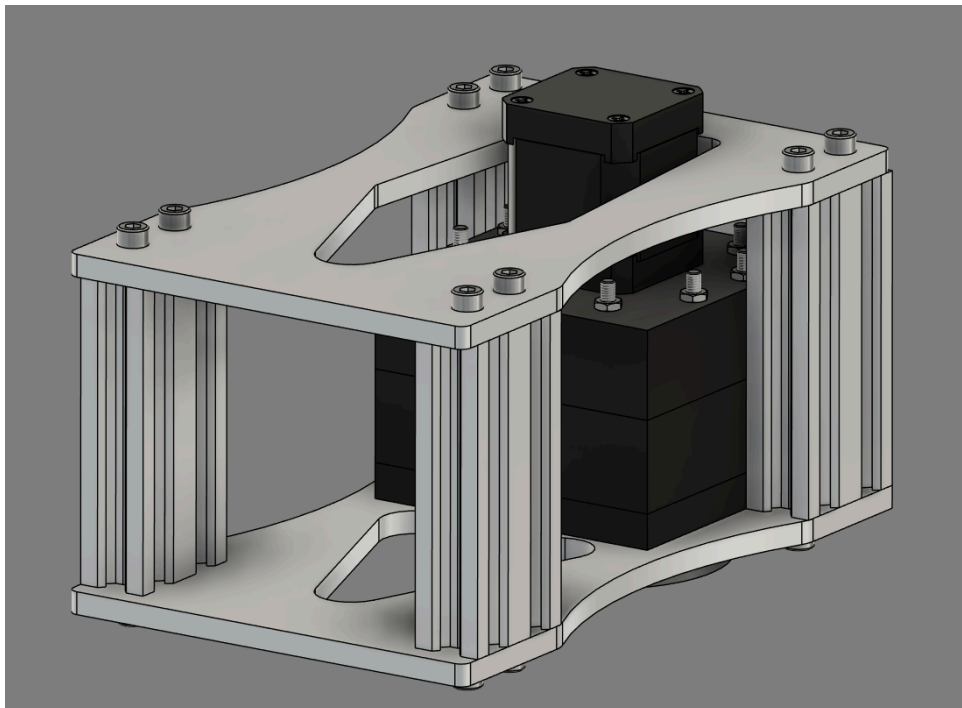
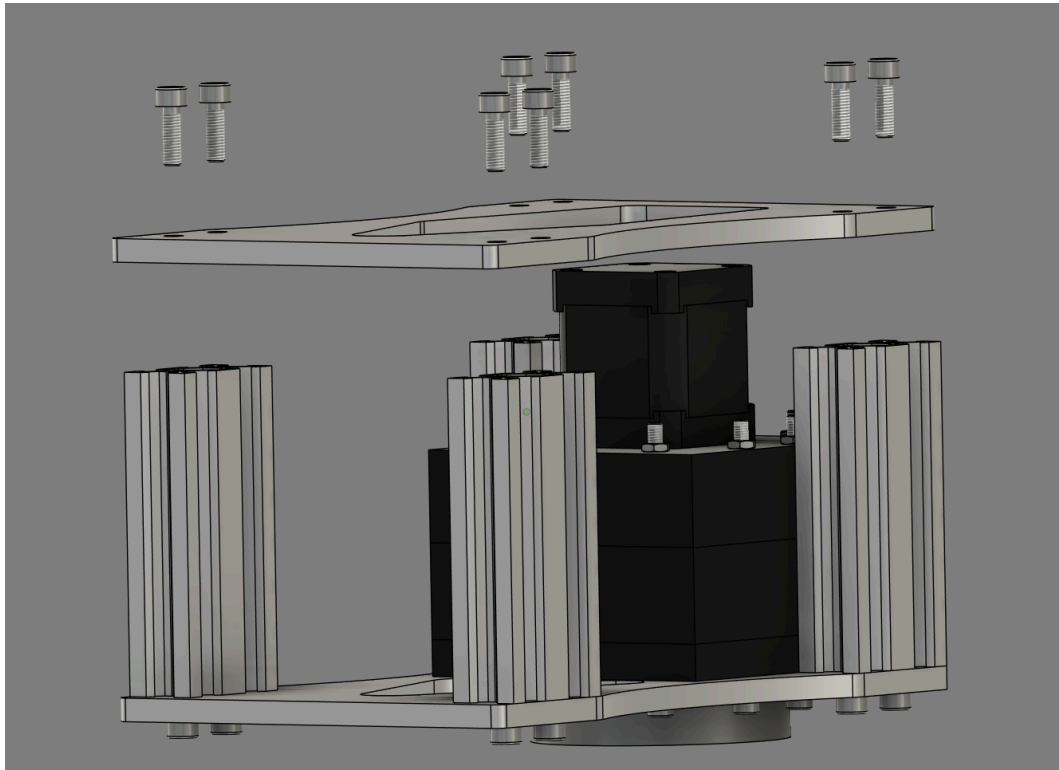
Step 1: Install 87.3mm 2040 Extrusions to Front Plate

- The equatorial mount frame has 4x 87.3mm 2040 extrusions. These connect the Front and Back Plates. We will first be mounting them to the Front Plate with 8x M5x16mm Bolts.



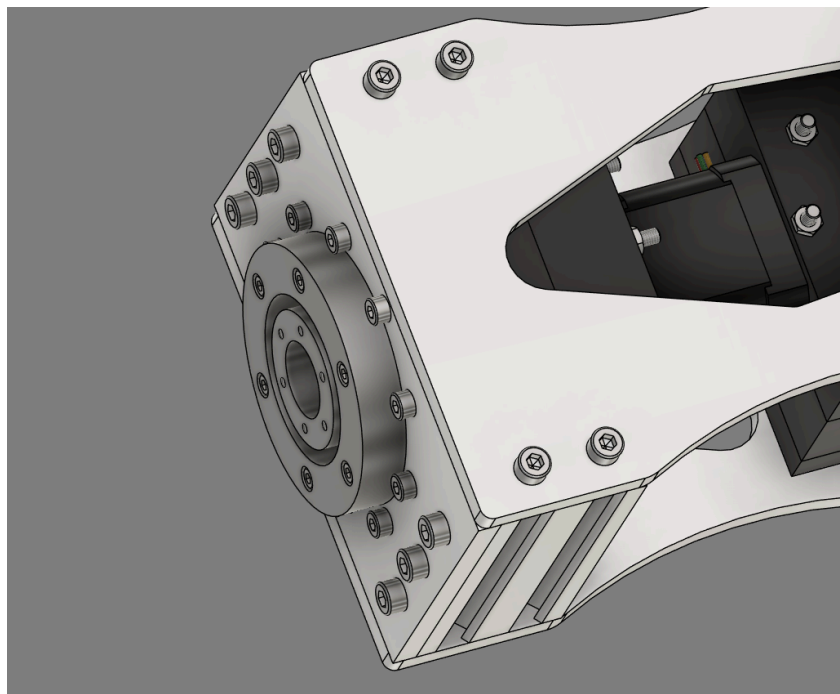
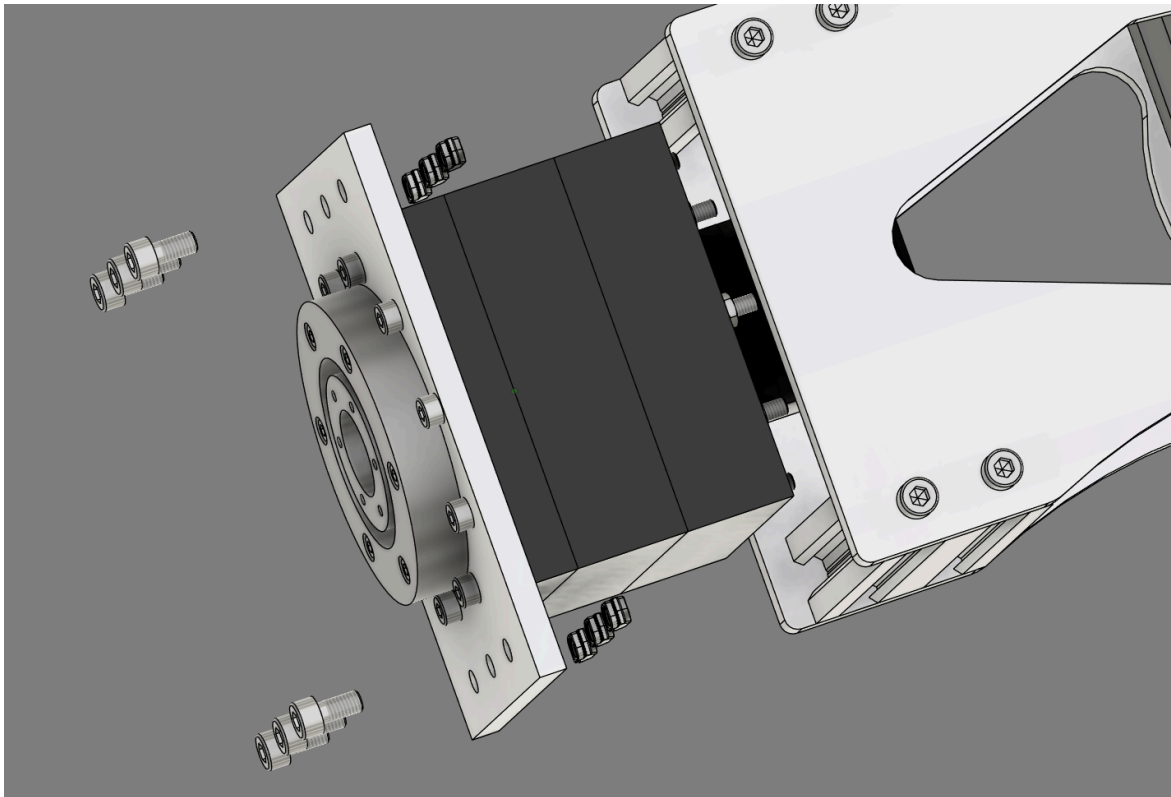
Step 2: Install 87.3mm 2040 Extrusions to Back Plate

- The 4x 87.3mm 2040 Extrusions connect to the Back Plate with 8x M5x16mm bolts.



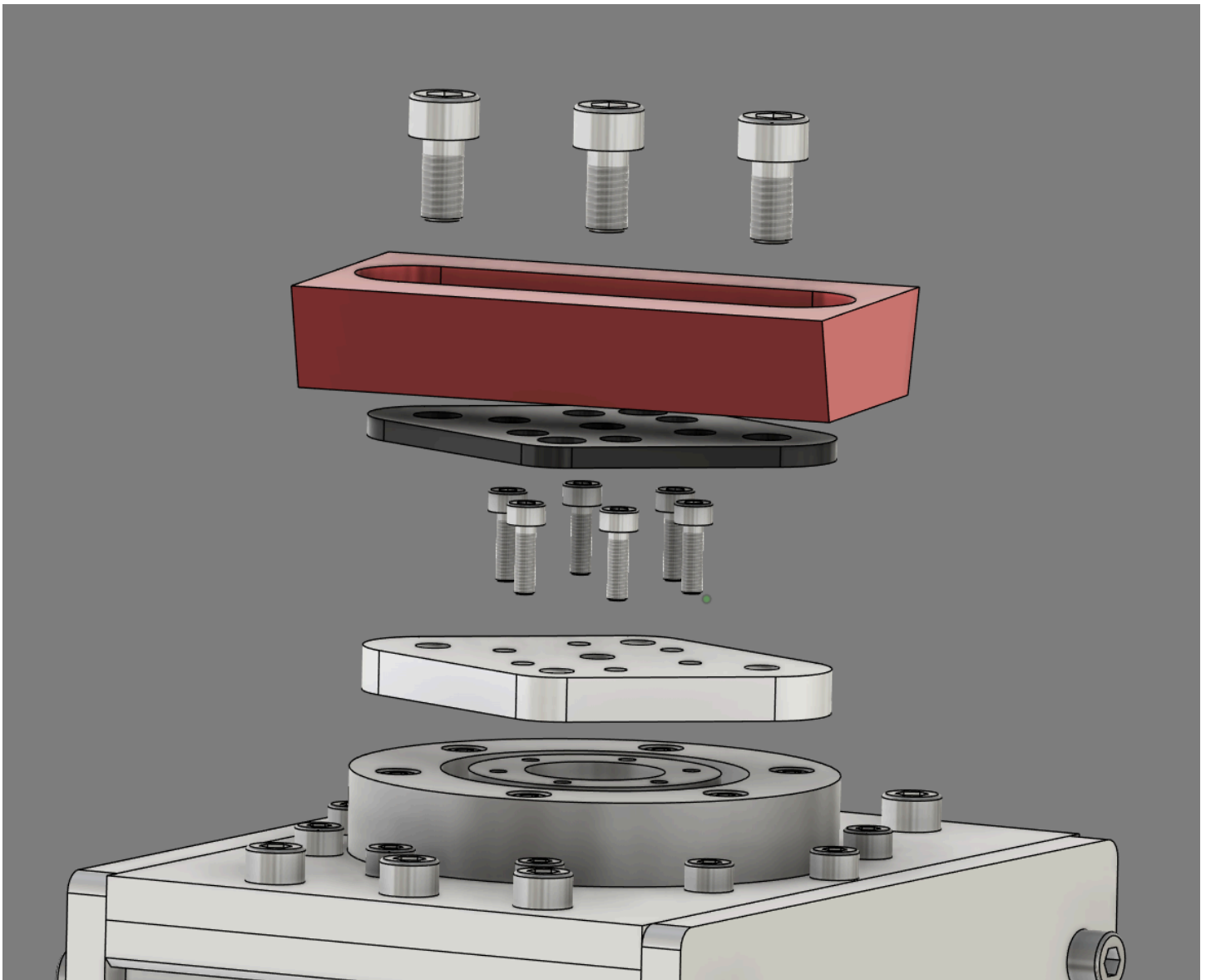
Step 3: Attach Bottom Plate 87.3mm 2040 Extrusions

- The Bottom Plate, with the R. A. Gearbox attached to it, mounts to the bottom 87.3mm 2040 Extrusions via 6x M5x10mm Bolts and 6x M5 T Nuts.



Step 4: Attach Mount Vixen Rail with Adapter Plate

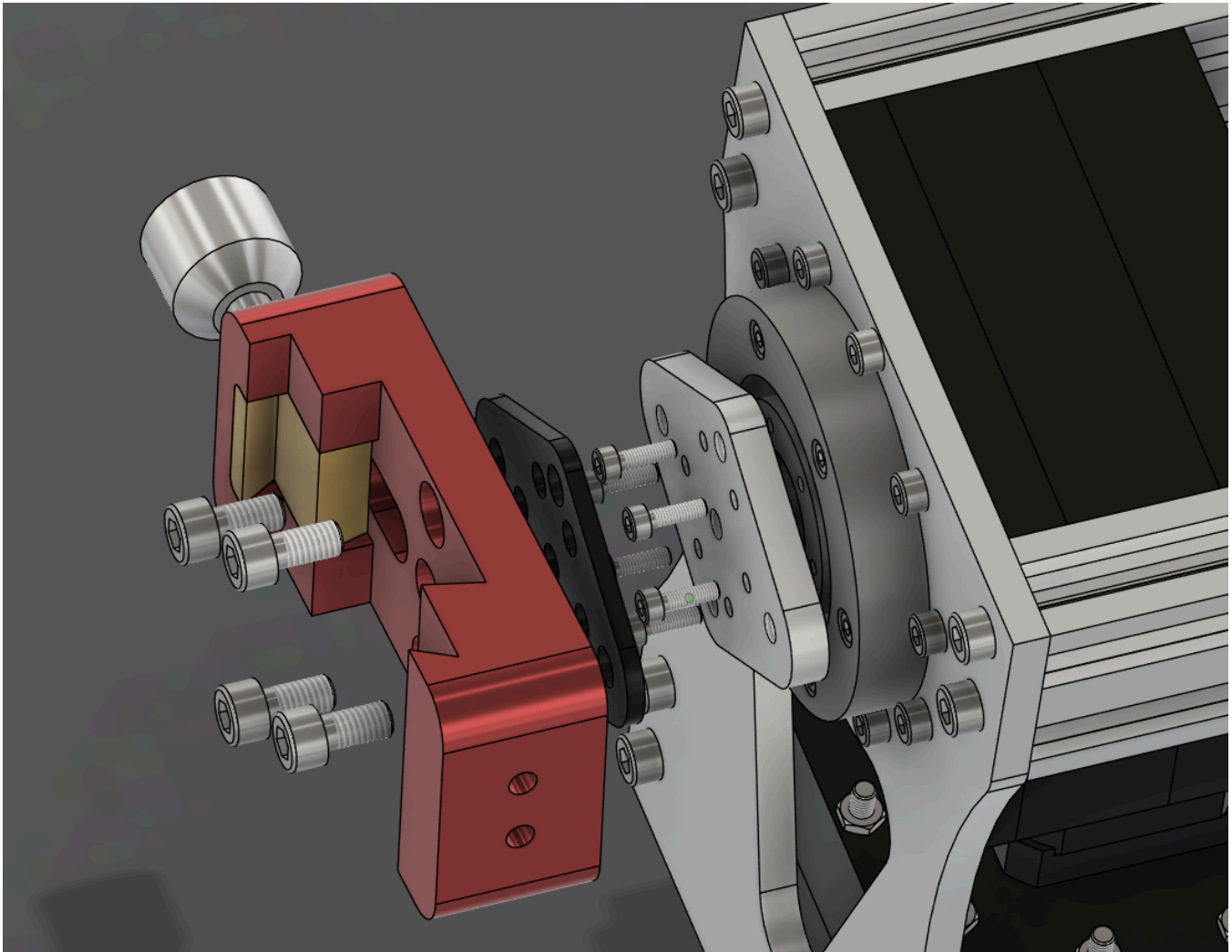
- For mounting the whole equatorial mount, the StarMount EQ has an 80mm Vixen Rail bolted to the base. This can be made longer, but I don't believe it is needed. This requires an adapter plate and adapter spacer so that it can rigidly attach to the output of the R. A. Gearbox RU42UCC0 bearing.



- The Adapter Plate is secured to the bearing output using 6x M3x10mm Bolts. The Vixen Rail then connects to the Adapter Plate through a printed Spacer via 3x M6x12mm Bolts.

Step 5: Attach Output Vixen Clamp and Adapter Plate

- For mounting equipment to the equatorial mount, the StarMount EQ has a Vixen Clamp bolted to the Dec Output. This requires an adapter plate and adapter spacer so that it can rigidly attach to the output of the Dec Gearbox RU42UUC0 bearing.



- The Adapter Plate is secured to the bearing output using 6x M3x10mm Bolts. The Vixen Clamp then connects to the Adapter Plate through a printed Spacer via 3x M6x12mm Bolts.

With that, your SpaceMount Equatorial Mount is now
Complete!